



CHAPTER

13

Thermodynamics

Sr. No.	Concept	Formulas	Other information
1.	First law of thermodynamics	According to 1 st law; $dQ = dU + dW$ or $dQ = dU + PdV$.	dW = external work. dQ = heat given to a system. dU = change in internal energy.
2.	Change in internal energy	$dU = nC_V (T_2 - T_1)$	n = number of moles. C_V = molar specific heat at constant volume T_1, T_2 = initial and final temperature of the system.
3.	Relation between two specific heats (Mayer's formula)	Heat lost or gained by n moles of a gas, (i) $\Delta Q = nC_p \Delta T$ (At constant pressure) (ii) $\Delta Q = nC_v \Delta T$ (At constant volume) Relation between them; $C_p - C_v = R$	C_p = molar specific heat at constant pressure. C_v = molar specific heat at constant volume. ΔT = change in temperature. R = universal gas constant. n = number of moles.
4.	Equation of state	Ideal gas equation can be written as; $PV = nRT$	P = pressure. V = volume. R = universal gas constant. T = temperature. n = number of moles.
5.	Isothermal process (temperature constant)	For ideal gas; $PV = \text{constant}$ or $P_1V_1 = P_2V_2$ For this process, $\Delta U = 0$, From first law of thermodynamics $dQ = dW$	P = pressure of the gas. V = volume of the gas.

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6.	Adiabatic process	(i) $PV^\gamma = \text{constant}$ or $P_1V_1^\gamma = P_2V_2^\gamma$ (ii) $TV^{\gamma-1} = \text{constant}$ or $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$ (iii) $\frac{P^{\gamma-1}}{T^\gamma} = \text{constant}$ or $\frac{P_1^{\gamma-1}}{T_1^\gamma} = \frac{P_2^{\gamma-1}}{T_2^\gamma}$ Where $\gamma = C_p / C_v$. For this process; $dQ = 0$ From first law thermodynamics, $dU = -dW$	γ = ratio of the specific heats. T_1, T_2 = initial and final temperature of the gas. V_1, V_2 = initial and final volume of the gas. P_1, P_2 = initial and final pressure of the gas.
7.	Isobaric process (pressure constant)	For ideal gas; $\frac{V}{T} = \text{constant}$ or $\frac{V_1}{T_1} = \frac{V_2}{T_2}$	T = temperature of the gas. V = volume of the gas.
8.	Isochoric process (volume constant)	For ideal gas; $\frac{P}{T} = \text{constant}$ or $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ For this process; $dW = 0$, From first law of thermodynamics $dQ = dU$	P = pressure of the gas. T = temperature of the gas.
9.	Cyclic process	In cyclic process, $dU = 0$ From first law of thermodynamics $dQ = dW$	dU = change in internal energy.
10.	Work done during the expansion of a gas	Work done is; $W = \int_{V_1}^{V_2} PdV$ = Area enclosed by the P - V curve	dV = change in volume. V_1 = initial volume. V_2 = final volume.
11.	Slope of isothermal and adiabatic curve	For isothermal curve; $\frac{dP}{dV} = \frac{-P}{V}$ For adiabatic curve; $\frac{dP}{dV} = \frac{-\gamma P}{V}$ $\left(\frac{dP}{dV}\right)_{\text{Adiabatic}} = \gamma \left(\frac{dP}{dV}\right)_{\text{isothermal}}$	γ = ratio of the specific heats of gases. P = pressure of the gas V = volume of the gas

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12.	Work done in different thermodynamic process	For isothermal process; $W_1 = 2.303 RT \log \frac{V_2}{V_1} = 2.303 RT \log \frac{P_1}{P_2}.$ For adiabatic process; $W_2 = \frac{R}{\gamma - 1} [T_1 - T_2] = \frac{1}{\gamma - 1} [P_1 V_1 - P_2 V_2]$ For isobaric process; $W_3 = P(V_2 - V_1) = nR (T_2 - T_1)$	T_2 = final temperature. T_1 = initial temperature. γ = ratio of specific heats.
13.	Polytropic process	$PV^n = \text{constant}$ $C = C_V + \frac{R}{1 - n}$	n = polytropic index. C = molar specific heat of gas. C_V = molar specific heat at constant volume.
14.	Efficiency of a heat engine	$\eta = \frac{\text{Work output}}{\text{Heat input}} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$ The efficiency of Carnot engine is given by $\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$	T_1, T_2 = the temperatures of source and sink respectively. η = efficiency of the engine. Q_1 = amount of heat absorbed per cycle from the source. Q_2 = amount of heat rejected per cycle to the sink.
15.	Refrigerator	$Q_1 = Q_2 + W$ Coefficient of performance of a refrigerator, $\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$ Relationship between η and β $\beta = \frac{1 - \eta}{\eta}$	Q_2 = heat drawn per cycle from sink Q_1 = heat rejected to source W = work done per cycle on refrigerator. T_1, T_2 = the temperatures of source and sink respectively
16.	C_P and C_V for mixture of gases.	C_P for a mixture of gas; $C_P = \frac{n_1 C_{P_1} + n_2 C_{P_2}}{n_1 + n_2}$ C_V for mixture of gas; $C_P = \frac{n_1 C_{V_1} + n_2 C_{V_2}}{n_1 + n_2}$	C_{P_1}, C_{P_2} = specific heat of two gases at constant pressure. C_{V_1}, C_{V_2} = specific heat of two gases at constant volume. n_1, n_2 = number of moles of two gases.